DOMAIN: PAINTER'S ALGORITHM · DEPTH-BASED RENDERING

COMPUTER GRAPHICS E1CSA 377

3D Building Visualization Using Depth-Based Rendering

Real-Time Application Project | Painter's Algorithm: sort far to near, paint in that order, GPU Z-buffer off

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FIG. 1 Entrance view, CPU sort ON, Z-buffer disabled. All 103,718 polygons painted far to near; nearer surfaces overwrite farther ones.



FIG. 5 Inside the hall, looking at the Ganesha statue. Columns, statue, plinth and courtyard are ordered correctly by the CPU sort alone.

103,718
polygons in the scene

1,584
objects sorted each frame

≈ 1 ms
CPU sort time (HUD: 0.45–1.10 ms)

Blender 5.2
LTS, host application

SORT OFF · Z OFF

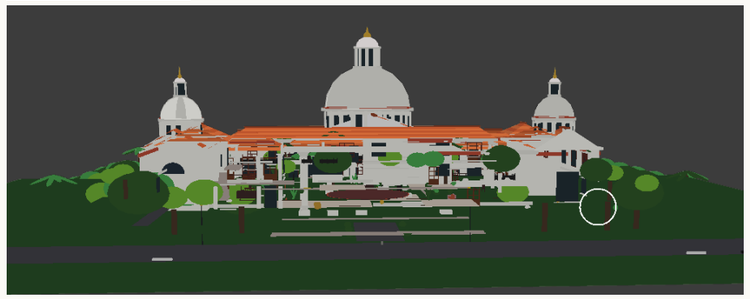


FIG. 2 Order scrambled: walls, trees and ground land in the wrong order and the interior shows through the facade.

SORT ON · Z OFF



FIG. 3 Painter's Algorithm only. Far-first sorting restores the facade: portico over wings, trees over lawn.

SORT ON · Z-BUFFER ON



FIG. 4 GPU depth test on as well. Almost identical, but the dome ribs missing in Fig. 3 are intact.

01 Paint the far things first



1ST Dome
z = 143.17 m

THEN Portico
z = 120.71 m

LAST Tree
z = 118.43 m

1. Measure each polygon's depth from the camera.
2. Sort the list, farthest first.
3. Draw in that order; nearer paint covers farther.

```
polys.sort(key=depth,
           reverse=True) # far → near
for p in polys: draw(p)
```

OBJECTIVE

To use depth sorting for rendering a complex building model.

EXPECTED OUTCOME

A building visualization demonstrating polygon ordering and visibility.

OBJECT DEPTH, FIG. 5 CAMERA (M)

Portico Pediment Front	11.88
Ganesha Belly	13.83
Grand Stair Step 1	16.80
Central Main Dome	19.12
Pavilion Left Core	26.03

PAINTER'S VS Z-BUFFER

Resolves visibility	**per polygon vs per pixel**
Cost	**a sort vs depth memory**
Weak spot	**cycles vs none**

CONCLUSION

Depth ordering decides what is visible. Sorting far to near rebuilt a 103,718-polygon building with the GPU depth buffer off; a per-pixel depth test still wins on fine detail.

BUILT WITH

Blender 5.2 LTS Python